Anwar Ibrahim

Machine Learning Researcher & Engineer

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HSE University profile

Professional Summary

Machine learning researcher and engineer developing reinforcement-learning methods and research software for accelerator beamline control. Experience spans continuous-control training, simulation environments, comparison with classical optimization, production-oriented machine learning, and agentic LLM systems. Author of peer-reviewed research and experienced in teaching generative models and practical machine learning.

Experience

Junior Research Fellow **Oct 2023–Present**
HSE University — LAMBDA *Moscow, Russia*

- Develop reinforcement-learning methods and research software for accelerator beamline control.
- Build Gymnasium-compatible simulation environments that connect learning agents with Elegant and SDDS accelerator tools.
- Train continuous-control agents, including DDPG, SAC, and TD3, and compare them with differential-evolution and Bayesian-optimization baselines.
- Publish and present research on simulation-based accelerator control in peer-reviewed journals and international scientific venues.

Lecturer — Generative Models in Machine Learning **Jan 2025–Present**
HSE University *Moscow, Russia*

- Teach *Generative Models in Machine Learning* in English, covering variational autoencoders, GANs, autoregressive models, and normalizing flows.
- Contribute practical machine-learning sessions and a reinforcement-learning competition to the All-Russian High-Energy Physics School.

AI Engineer — Agentic Systems **Dec 2025–Oct 2026**
Eolos AI

- Developed components for private agentic AI systems built around large language models and multi-step LangGraph workflows.
- Implemented human-in-the-loop review and feedback paths, managed application data with Convex, and used LangSmith for tracing and evaluation.
- Worked on the orchestration, observability, and reliability of LLM-based workflows.

Machine Learning Engineer **Jun 2020–Sep 2023**
EXPentech *Damascus, Syria*

- Developed and evaluated machine-learning models, prepared and explored datasets, and supported model deployment and monitoring.

Education

PhD Candidate, Artificial Intelligence and Machine Learning **Nov 2023–Present**
HSE University *Moscow, Russia*

- Doctoral research: reinforcement-learning-based control of accelerator beamlines.

Master's in Data Science **2021–Jul 2023**
HSE University *Moscow, Russia*

- Degree programme in Applied Mathematics and Information Science.

Bachelor's in Computer Software Engineering **2015–2020**
Higher Institute for Applied Sciences and Technology *Damascus, Syria*

Peer-Reviewed Publications

1. **Anwar Ibrahim**, Fedor Ratnikov, Maxim Kaledin, Alexey Petrenko, Denis Derkach. *RL-ABC: Reinforcement Learning for Accelerator Beamline Control.*
   Computer Physics Communications, 329, 110407 (2026).
2. **Anwar Ibrahim**, Alexey Petrenko, Maxim Kaledin, Ehab Suleiman, Fedor Ratnikov, Denis Derkach. *Reinforcement Learning for Accelerator Beamline Control: A Simulation-Based Approach.*
   International Journal of Modern Physics E, 35(07) (2026).
3. **Anwar Ibrahim**, Denis Derkach, Alexey Petrenko, Fedor Ratnikov, Maxim Kaledin. *Optimization of the Accelerator Control by Reinforcement Learning: A Simulation-Based Approach.*
   Physics of Particles and Nuclei, 56(6), 1476–1481 (2025).

Selected Conference Contributions

- **CHEP 2026, Bangkok — Talk:** RL-ABC: Reinforcement Learning for Accelerator Beamline Control.
- **ACAT 2025 — Oral contribution:** Optimisation of the Accelerator Control by Reinforcement Learning: A Simulation-Based Approach.
- **NUCLEUS 2025 — Talk:** Reinforcement Learning for Accelerator Beamline Control.
- **CERN Workshop 2025:** Reinforcement Learning for Accelerator Beamline Control.
- **MMCP 2024:** Optimization of Accelerator Control by Reinforcement Learning.

Selected Projects

RL-ABC Repository
Research framework that converts Elegant accelerator lattice configurations into reinforcement-learning environments, with continuous-control agents and classical optimization baselines.

Implicit Q-Learning on Hopper Repository
PyTorch implementation of Implicit Q-Learning on the D4RL Hopper expert dataset, including actor, value, twin-critic, and target networks.

Interpretable Reinforcement Learning for Pong Repository
Analysis toolkit using saliency maps, Q-value inspection, paddle-position analysis, and an interactive dashboard.

Multilingual Contexto Repository
Semantic word-guessing game built with BERT-family models and Streamlit, supporting Arabic, English, and Russian.

Technical Skills

Machine Learning	Reinforcement learning, generative models, diffusion models, NLP, computer vision, human-in-the-loop systems
Programming	Python, C#, SQL
Frameworks	PyTorch, TensorFlow, Gymnasium, Stable-Baselines3, Streamlit, OpenCV
Agentic AI	LangChain, LangGraph, LangSmith, Convex, LLM workflow orchestration
Scientific Tools	Elegant, SDDS, simulation-based optimization, experiment tracking
Engineering	Git, model evaluation, deployment, monitoring
Languages	Arabic, English, Russian

Selected Research Coverage

HSE University science feature on RL-ABC and accelerator-control research (2026), with additional coverage in Gazeta.Ru, Vedomosti, Mail.ru Science, Naked Science, CNews, and other outlets.